

# Chencheng Liang

PhD in Computer Science

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## Research Experience

2019-01 – 2025-06

### Doctoral Researcher in Machine Learning for Automated Reasoning

Uppsala University, Sweden

- Developed a GNN-based framework for constraint solving, Horn-clause solving, and word-equation solving, including dataset generation, graph representation, model training, solver integration, and empirical evaluation in Python and PyTorch
- Ran large-scale GPU experiments on Slurm-managed HPC clusters, automating batch submission, hyperparameter sweeps, checkpointing, logging, and result aggregation across repeated runs
- Built modular and reproducible experimental workflows for comparing model behaviour, solver performance, and generalisation across different problem classes
- Collaborated with researchers in machine learning and automated reasoning, supervised master's projects, and translated research results into peer-reviewed publications and international presentations

## Work Experience

2026-06 – Present

### Lecturer in Uppsala University

- Lecturer for the course Artificial Intelligence (1DL340), an English-taught Master's level course at Uppsala University.
- Responsible for teaching, assessment, and course administration throughout the autumn 2026 teaching period. delivery

2025-08 – 2026-06

### Independent AI & Automation Consultant

- Built and operated Python-based AI workflows for live e-commerce and consulting projects, combining web scraping, image-based information extraction, structured LLM outputs, schema-based data processing, and multilingual normalization
- Developed backend workflows for an independent Shopify store covering product ingestion, attribute extraction, categorization, SEO metadata, localization, and marketing-content generation, converting manual operating steps into repeatable automated pipelines
- Worked directly with business owners to break operational requirements into technical implementation tasks, select and integrate tools, and resolve data, model-output, workflow, and deployment issues during delivery

## Skills

- **Programming:** Python, C/C++, Java, Scala, Bash
- **AI/ML:** PyTorch, TensorFlow, PyG, GNNs, Transformers, RNNs, Reinforcement Learning, NLP
- **Model Research:** Dataset generation, model training, hyperparameter tuning, checkpointing, experiment tracking, empirical evaluation
- **LLM & Automation:** Structured LLM outputs, schema-based extraction, prompt design, tool and API integration, workflow automation
- **Compute & Systems:** GPU/HPC environments, Slurm, NAISS, Linux/Unix, Docker, Git, MLflow, AWS
- **Backend & Data:** Python backend services, REST APIs, web scraping, data pipelines, image-based extraction, multilingual data standardization

- **Automated Reasoning:** Formal verification, constraint solving, Horn clauses, word equations, algorithm design
- **Languages:** Chinese (Native), English (Fluent), Japanese (Basic)

## Education

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| 2019 – 2025 | <b>Uppsala University, Sweden</b><br><i>PhD in Computer Science</i><br>Thesis: “Learning to Guide Automated Reasoning: A GNN-Based Framework”<br>Advisors: Philipp Rümmer, Yi Wang, Marc Brockschmidt<br>Opponent: Stephan Schulz |
| 2016 – 2018 | <b>University of Göttingen, Germany</b><br><i>Master of Science in Computer Science</i><br>Thesis: “Topology Control Using Fuzzy Game Theory in Mobile Underwater Sensor Networks”<br>Grade: 1.0 (Sehr gut, highest distinction)  |
| 2010 – 2014 | <b>Chengdu University, China</b><br><i>Bachelor of Science in Software Engineering</i><br>Graduated 1st in class                                                                                                                  |

## Selected Applied AI Projects

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| 2025 | <b>CozySteps - LLM-powered Activity Recommender</b><br><i>Impact Hack 2025, AI Sweden</i> <ul style="list-style-type: none"><li>• Built mobile app prototype for families with autistic children</li><li>• Implemented LLM-based activity recommendation engine</li><li>• Won recognition for addressing real societal challenges</li></ul> |
| 2023 | <b>Insilico - AI-Driven Art Installation</b><br><i>Uppsala Art Museum</i> <ul style="list-style-type: none"><li>• Integrated AI-based sentiment analysis for Mat Collishaw’s art installation</li><li>• Developed real-time Twitter sentiment analysis system</li><li>• Connected digital hostility to physical art expression</li></ul>    |

## Teaching Experience

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| 2019 – 2025 | <b>Uppsala University, Sweden</b><br><i>Course Instructor &amp; Teaching Assistant</i> <ul style="list-style-type: none"><li>• <b>Natural Computation Methods for Machine Learning</b> (Master’s, 10 credits, 2019-2025)</li><li>• <b>Imperative and Object-Oriented Programming Methodology</b> (Bachelor’s, 10 credits, 2023-2024)</li><li>• <b>Project CS</b> (Master’s project course, 30 credits, 2022-2024)</li><li>• <b>Introduction to Studies in Embedded Systems</b> (Master’s, 5 credits, 2019-2021)</li></ul> |
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## Thesis & Project Supervision

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| 2024 | <b>Federated Learning meets Large Language Models</b><br><i>Master's Project Supervision</i><br>Students: Mahtab Mirhaj, Oguzhan Ersoy, Yixin Huang, Shiqi Shu<br>Ericsson Research & Uppsala University                                                                                                           |
| 2023 | <b>Distributed Mobility Prediction in Telecommunication Networks</b><br><i>Master's Project Supervision</i><br>Students: Shreyansh Singh, Lifang Zheng, Savvas Giortsis, Tobias Lass, Yilei Zheng, Salum Nassor<br>Ericsson Research & Uppsala University                                                          |
| 2022 | <b>Shortest Path Reliable Estimation: Applying Graph Neural Networks - SpreadNet</b><br><i>Master's Project Supervision</i><br>Students: ChanVuth Chea, Boli Gao, Jennifer Gross, Haouyuan Li, Paarth Sanhotra, George-Alexandru Stoian, Sofia Afnaan Syed, Haodong Zhao<br>Ericsson Research & Uppsala University |
| 2021 | <b>Exploring Properties and Limitations of Graph Neural Networks (GNNs) in Software Verification</b><br><i>Master's Thesis Supervision</i><br>Student: Kexin Xu<br>Uppsala University                                                                                                                              |

## Professional Service

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- **Reviewer for Conferences:**
  - FMCAD 2025, FM 2024, LPAR 2024, VMCAI 2024, FMCAD 2023, LPAR 2023
  - FMCAD 2022, IJCAR 2022, FroCoS 2021, CADE 2021, TACAS 2020
  - iFM 2019, FroCoS 2019, CADE 2019
- **Talks & Presentations:**
  - “Guiding Constraint Horn Clauses Solving using Graph Neural Networks” - ISTA, Austria, 2024
  - “Boosting Constrained Horn Solving by Unsat Core Learning” - HCVS Workshop, Luxembourg, 2024
  - “Exploring Representation of Horn Clauses using GNNs” - AITP Conference, France, 2022
  - “Guiding Interpolation for Model Checking by Deep Learning Techniques” - AVM, Czech Republic, 2019

## Publications

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| 2025 | 1. | Abdulla, P. A., Atig, M. F., Cailler, J., <b>Liang, C.</b> & Rümmer, P. When GNNs Met a Word Equations Solver: Learning to Rank Equations. In <i>15th International Symposium on Frontiers of Combining Systems (FroCoS)</i> (Springer, 2025).                                            |
| 2024 | 2. | Abdulla, P. A., Atig, M. F., Cailler, J., <b>Liang, C.</b> & Rümmer, P. Guiding Word Equation Solving Using Graph Neural Networks. In <i>22nd International Symposium on Automated Technology for Verification and Analysis (ATVA)</i> (Springer, 2024).                                  |
|      | 3. | Abdulla, P. A., <b>Liang, C.</b> & Rümmer, P. Boosting Constrained Horn Solving by Unsat Core Learning. In <i>25th International Conference on Verification, Model Checking, and Abstract Interpretation (VMCAI)</i> (Springer, 2024).                                                    |
| 2022 | 4. | <b>Liang, C.</b> , Rümmer, P. & Brockschmidt, M. Exploring Representation of Horn clauses using GNNs. In <i>8th Workshop on Practical Aspects of Automated Reasoning (PAAR)</i> (2022).                                                                                                   |
| 2021 | 5. | Yuan, Y., <b>Liang, C.</b> , Chen, X., Baker, T. & Fu, X. Adaptive Fuzzy Game-Based Energy-Efficient Localization in 3D Underwater Sensor Networks. <i>ACM Transactions on Internet Technology</i> <b>22</b> , 1–20 (2021).                                                               |
| 2019 | 6. | Yuan, Y., <b>Liang, C.</b> , Kaneko, M., Chen, X. & Hogrefe, D. Topology Control for Energy-Efficient Localization in Mobile Underwater Sensor Networks Using Stackelberg Game. <i>IEEE Transactions on Vehicular Technology</i> <b>68</b> , 1487–1500 (2019).                            |
| 2017 | 7. | Yang, Y., <b>Liang, C.</b> & Ji, S. Comments on “Fuzzy multicriteria decision making method based on the improved accuracy function for interval-valued intuitionistic fuzzy sets” by Ridvan Sahin. <i>Soft Computing</i> <b>21</b> , 3033–3035 (2017).                                   |
| 2015 | 8. | Yang, Y., <b>Liang, C.</b> , Ji, S. & Liu, T. Adjustable soft discernibility matrix based on picture fuzzy soft sets and its applications in decision making. <i>Journal of Intelligent &amp; Fuzzy Systems: Applications in Engineering and Technology</i> <b>29</b> , 1711–1722 (2015). |